

**Exp. No:01****Date:****CLASSES AND OBJECTS**

**PROGRAM :**

```
import java.util.*;

class Box{

    double w,h,d;

    Box(){

        w=2;

        h=2;

        d=2;

    }

    Box(double a){

        w=a;

        h=2;

        d=3;

    }

    Box(double a,double b){

        w=a;

        h=b;

        d=3;

    }

    Box(double a,double b,double c){

        w=a;

        h=b;

        d=c;

    }

    double Volume(){

        return w*h*d;

    }

}
```

```
}  
  
class Vol_Box{  
  
    public static void main(String args[]) {  
  
        Scanner s=new Scanner(System.in);  
  
        System.out.println("\t Classes and Objects");  
  
        double a1,b1,b2,c1,c2,c3;  
  
        System.out.println("Enter One value :");  
  
        a1=s.nextDouble();  
  
        System.out.println("Enter Two values :");  
  
        b1=s.nextDouble();  
  
        b2=s.nextDouble();  
  
        System.out.println("Enter Three values :");  
  
        c1=s.nextDouble();  
  
        c2=s.nextDouble();  
  
        c3=s.nextDouble();  
  
        Box box=new Box();  
  
        Box box1=new Box(a1);  
  
        Box box2=new Box(b1,b2);  
  
        Box box3=new Box(c1,c2,c3);  
  
        System.out.println("The Volume of Box1 is :"+box.Volume());  
  
        System.out.println("The Volume of Box2 is :"+box1.Volume());  
  
        System.out.println("The Volume of Box3 is :"+box2.Volume());  
  
        System.out.println("The Volume of Box4 is :"+box3.Volume());  
  
    }  
  
}
```

**OUTPUT :**

```
ragupathi@ragu-2k05:~/Msc_IT_labs/Adavance_Java_Lab/Ex01$ java Vol_Box
Classes and Objects
Enter One value :
2
Enter Two values :
2
3
Enter Three values :
4
5
6
The Volume of Box1 is :8.0
The Volume of Box2 is :12.0
The Volume of Box3 is :18.0
The Volume of Box4 is :120.0
```



**Exp. No:02****Date:****CLASSES AND OBJECTS**

**PROGRAM :**

```
import java.util.*;

class Area_Shapes{

    double pi=3.14;

    void area(int a){

        System.out.println("\nThe Area of the Square is :"+a*a);

    }

    void area(int a,int b){

        System.out.println("The Area of the Rectangle is :"+a*b);

    }

    double area(double r){

        return pi*r*r;

    }

}

class Shapes{

    public static void main(String args[]){

        System.out.println("\t Implementing Classes");

        Scanner s=new Scanner(System.in);

        double c,area_cir;

        int sq,l,b;

        Area_Shapes ar=new Area_Shapes();

        System.out.print("\nEnter the Side for Square :");

        sq=s.nextInt();

        System.out.print("Enter the Length of the Rectangle :");

        l=s.nextInt();

        System.out.print("Enter the Breadth of the Rectangle :");

        b=s.nextInt();
```

```
System.out.print("Enter the Radius of the Circle :");  
c=s.nextDouble();  
ar.area(sq);  
ar.area(l,b);  
area_cir=ar.area(c);  
System.out.println("The Area of the Circle is :"+area_cir);  
}  
}
```

**OUTPUT :**



**Exp. No: 03****Date:****STRINGS**

**PROGRAM :**

```
import Bank.Account;

import java.util.*;

class User{

    public static void main(String[] args) {

        Scanner s= new Scanner(System.in);

        Account ac=new Account();

        System.out.print("\nEnter the Name :");

        String name=s.nextLine();

        System.out.print("\nEnter the Initial Amount :");

        int amount=s.nextInt();

        s.nextLine();

        ac.createAccount(name, amount);

        System.out.println("""

            1)Credit

            2)withdraw

            3)Balance

            4)Exit

            """);

        while(true){

            System.out.print("\nEnter What you want to do :");

            String msg=s.nextLine().toLowerCase();

            if("credit".equals(msg)){
```

```
        System.out.print("\nEnter the amount :");
        int amt=s.nextInt();
        ac.credit(amt);
        s.nextLine();
    }
    else if("withdraw".equals(msg)){
        System.out.print("\nEnter the amount :");
        int amt=s.nextInt();
        ac.withdraw(amt);
        s.nextLine();
    }
    else if("balance".equals(msg)){
        ac.disp_balance();
    }
    else if("exit".equals(msg)){
        break;
    }
    else{
        System.out.println("Invalid Action !");
    }
}
}
```

**OUTPUT :**



**Exp. No: 04****Date:****COLLECTION**

**PROGRAM :**

```
import java.util.*;

class Collection{

    public static void main(String args[]) {

        Scanner s=new Scanner(System.in);

        ArrayList<Integer> al=new ArrayList<>();

        System.err.println("Enter Five Numbers :");

        for(int i=1;i<=5;i++){

            System.out.print("Enter the Element "+i+" :");

            al.add(s.nextInt());

        }

        int sum=0;

        for (int a:al){

            sum+=a;

        }

        System.out.println("The Sum of the Elements is :"+sum);

        LinkedList<String> l=new LinkedList<>();

        HashSet<String> hs=new HashSet<>();

        TreeSet<String> ts=new TreeSet<>();

        System.out.println("\nEnter Five Letters :");

        s.nextLine();

        for(int i=1;i<=5;i++){

            System.out.print("Enter the Letter "+i+" :");
```

```
        String str=s.nextLine();

        l.add(str);

        hs.add(str);

        ts.add(str);

    }

    System.out.println("\nThe Linked List Elements are :"+l);

    System.out.print("Enter the Element to remove :");

    String s1=s.nextLine();

    System.out.print("Enter the Position to remove :");

    int s2=s.nextInt();

    System.out.println("Before Remove :"+l);

    l.remove(s1);

    l.remove(s2);

    System.out.println("After Remove :"+l);

    System.out.println("\nThe HashSet(Unordered) elements are :"+hs);

    System.out.println("\nThe TreeSet(Ordered) Elements are :"+ts);

}

}
```

**OUTPUT :**



**Exp. No: 05****Date:****DATE AND CALENDAR**

**PROGRAM :**

```
import java.util.*;

class Date_Time{

    public static void main(String[] args) {

        Scanner s=new Scanner(System.in);

        Calendar cal=Calendar.getInstance();

        Date dt=new Date();

        System.out.println("\t Date and Time" );

        System.out.println("\n\nThe Current Date,Time and Timezone in a System is :
"+dt);

        System.out.println("The Current Time in Milli Second is :"+dt.getTime());

        String
months[]={ "Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Oct", "Sep", "Nov", "Dec" };

        System.out.print("\nDate : ");

        System.out.print(months[cal.get(Calendar.MONTH)]);

        System.out.print(" " + cal.get(Calendar.DATE));

        System.out.print(", " + cal.get(Calendar.YEAR));

        System.out.print("\nTime : ");

        System.out.print(cal.get(Calendar.HOUR));

        System.out.print(" : " + cal.get(Calendar.MINUTE));

        System.out.print(" : " + cal.get(Calendar.SECOND));

        System.out.println("\n");
```

```
        System.out.print("Enter the Hour :");  
        int h=s.nextInt();  
        System.out.print("Enter the Minutes :");  
        int m=s.nextInt();  
        System.out.print("Enter the Seconds :");  
        int sc=s.nextInt();  
  
        cal.set(Calendar.HOUR,h);  
        cal.set(Calendar.MINUTE,m);  
        cal.set(Calendar.SECOND,sc);  
  
        System.out.print("\nUpdated Time :");  
        System.out.print(cal.get(Calendar.HOUR));  
        System.out.print(" : "+cal.get(Calendar.MINUTE));  
        System.out.print(" : "+cal.get(Calendar.SECOND)+"\n");  
    }  
}
```

**OUTPUT :**



**Exp. No: 06****Date:****PACKAGES**

**PROGRAM :****Account.java :**

```
package Bank;

public class Account {

    String name;

    int balance;

    public void createAccount(String name, int initialAmount) {

        this.name = name;

        this.balance = initialAmount;

    }

    public int credit(int amount){

        balance+=amount;

        System.out.println("The Amount Credited Successfully");

        return balance;

    }

    public int withdraw(int amount){

        if(amount<=balance){

            balance-=amount;

            System.out.println("The Amount Withdrawn Successfully");

        }

        else{

            System.out.println("Your balance is Low");

        }

        return balance;

    }

    public void disp_balance(){
```

```

        System.out.println("Name :"+name);
        System.out.println("Balance :"+balance);
    }
}

```

### **User.java :**

```

import Bank.Account;
import java.util.*;
class User{
    public static void main(String[] args) {
        Scanner s= new Scanner(System.in);
        Account ac=new Account();

        System.out.print("\nEnter the Name :");
        String name=s.nextLine();
        System.out.print("\nEnter the Initial Amount :");
        int amount=s.nextInt();
        s.nextLine();
        ac.createAccount(name, amount);

        System.out.println("""
            1)Credit
            2)withdraw
            3)Balance
            4)Exit
            """);
    }
}

```

```
while(true){  
    System.out.print("\nEnter What you want to do :");  
    String msg=s.nextLine().toLowerCase();  
    if("credit".equals(msg)){  
        System.out.print("\nEnter the amount :");  
        int amt=s.nextInt();  
        ac.credit(amt);  
        s.nextLine();  
    }  
    else if("withdraw".equals(msg)){  
        System.out.print("\nEnter the amount :");  
        int amt=s.nextInt();  
        ac.withdraw(amt);  
        s.nextLine();  
    }  
    else if("balance".equals(msg)){  
        ac.disp_balance();  
    }  
    else if("exit".equals(msg)){  
        break;  
    }  
    else{  
        System.out.println("Invalid Action !");  
    }  
}  
}
```

**OUTPUT :**



**Exp. No: 07****Date:****EXCEPTION HANDLING**

**PROGRAM :**

```
import java.util.*;

class Exception_Hand {

    public static void main(String[] args) {

        int a = args.length;

        Scanner s = new Scanner(System.in);

        System.out.println("Value of a is : " + a);

        try {

            if (a == 0) {

                int b = 24;

                a = b / (b - b);

            }

            if (a == 1) {

                int A[] = {};

                A[16] = 20;

            }

            if (a == 2) {

                System.out.print("Enter Array size: ");

                int n = s.nextInt();

                int B[] = new int[n];

                B[17] = 15;

            }

        } catch (ArithmeticException e) {

            System.out.println("Do not use zero as a divisor. " + e);

        } catch (ArrayIndexOutOfBoundsException e) {

            System.out.println("Array size exceeds. " + e);

        }

    }

}
```

```
    } catch (NegativeArraySizeException e) {  
        System.out.println("Array size cannot be negative value. " + e);  
    } finally {  
        System.out.println("Exceptions Completed.");  
    }  
}  
}
```

**OUTPUT :**



**Exp. No: 08****Date:****THREADS**

**PROGRAM :**

```
class Q {  
    int n;  
  
    boolean valueSet = false;  
  
    synchronized int get() {  
        if (!valueSet)  
            try {  
                wait();  
            } catch (InterruptedException e) {  
                System.out.println("InterruptedException caught");  
            }  
        System.out.println("Got: " + n);  
        valueSet = false;  
        notify();  
        return n;  
    }  
  
    synchronized void put(int n) {  
        if (valueSet)  
            try {  
                wait();  
            } catch (InterruptedException e) {  
                System.out.println("InterruptedException caught");  
            }  
        this.n = n;  
        valueSet = true;  
        System.out.println("Put: " + n);  
        notify();  
    }  
}
```

```
    }  
}  
  
class Producer implements Runnable {  
    Q q;  
    Producer(Q q) {  
        this.q = q;  
        new Thread(this, "Producer").start();  
    }  
    public void run() {  
        int i = 1;  
  
        while (i <= 10) {  
            q.put(i++);  
        }  
    }  
}  
  
class Consumer implements Runnable {  
    Q q;  
    Consumer(Q q) {  
        this.q = q;  
        new Thread(this, "Consumer").start();  
    }  
    public void run() {  
        while (true) {  
            q.get();  
        }  
    }  
}
```

```
}  
  
class PC {  
  
    public static void main(String args[]) {  
  
        System.out.println("Press Control-C to stop.");  
  
        Q q = new Q();  
  
        new Producer(q);  
  
        new Consumer(q);  
  
    }  
}
```

**OUTPUT :**



**Exp. No: 09****Date:****JDBC**

**PROGRAM :**

```
import java.sql.*;

public class Student {

    public static void main(String args[]) {

        try {

            // Load MySQL JDBC Driver

            Class.forName("com.mysql.cj.jdbc.Driver");

            // Database connection details

            String url = "jdbc:mysql://localhost:3306/ragu";

            String user = "root";

            String password = "ragu@123";

            // Establish connection

            Connection con = DriverManager.getConnection(url, user, password);

            System.out.println("Created DB Connection...");

            // Create Statement

            Statement stmt = con.createStatement();

            // Execute Query

            ResultSet rs = stmt.executeQuery("SELECT Id, name, class FROM student");

            // Print Column Names

            ResultSetMetaData rsmd = rs.getMetaData();

            int count = rsmd.getColumnCount();
```

```
        for (int i = 1; i <= count; i++) {  
            System.out.print(rsmd.getColumnName(i) + "\t");  
        }  
        System.out.println();  
  
        // Print Records  
        while (rs.next()) {  
            System.out.print(rs.getInt("id") + "\t");  
            System.out.print(rs.getString("name") + "\t");  
            System.out.print(rs.getString("class") + "\t");  
            System.out.println();  
        }  
  
        // Close Resources  
        rs.close();  
        stmt.close();  
        con.close();  
  
    } catch (ClassNotFoundException e) {  
        System.out.println("JDBC Driver Not Found!");  
        e.printStackTrace();  
    } catch (SQLException e) {  
        System.out.println("Database Connection Error!");  
        e.printStackTrace();  
    }  
}  
}
```

**OUTPUT :**



**Exp. No: 10****Date:****APPLETS**

**PROGRAM :**

```
import java.awt.*;

import java.applet.*;

/*
<applet code="SimpleBanner" width="700" height="50">
</applet>
*/

public class SimpleBanner extends Applet implements Runnable {

    String msg = " Accept Everything Nothing is Permanent.";

    Thread t = null;

    int state;

    boolean stopFlag;

    // Set colors and initialize thread

    public void init() {

        setBackground(Color.blue);

        setForeground(Color.white);

        setFont(new Font("Dialog", Font.PLAIN, 50));

    }

    // Start thread

    public void start() {

        t = new Thread(this);

        stopFlag = false;

        t.start();

    }

}
```

```
// Entry point for the thread that runs the banner

public void run() {

    char ch;

    // Display banner

    for (;;) {

        try {

            repaint();

            Thread.sleep(250);

            ch = msg.charAt(0);

            msg = msg.substring(1, msg.length());

            msg += ch;

            if (stopFlag)

                break;

        } catch (InterruptedException e) {

        }

    }

}

// Pause the banner

public void stop() {

    stopFlag = true;

    t = null;

}
```

```
// Display the banner  
public void paint(Graphics g) {  
    g.drawString(msg, 100, 300);  
}  
}
```

**OUTPUT :**



**Exp. No: 11****Date:****EVENT HANDLING**

**PROGRAM :**

```
import java.applet.*;

import java.awt.*;

import java.awt.event.*;

/*

<applet code="Event_Hand" width="1200" height="1200">

</applet>

*/

public class Event_Hand extends Applet implements ActionListener {

    TextField a, b;

    double sum;

    Button ab;

    public void init() {

        Label value1 = new Label("First Number : ", Label.RIGHT);

        Label value2 = new Label("Second Number : ", Label.RIGHT);

        a = new TextField(10);

        b = new TextField(10);

        ab = new Button("Add");
```

```
        add(value1);
        add(a);
        add(value2);
        add(b);
        add(ab);

        ab.addActionListener(this);
    }

    public void actionPerformed(ActionEvent ae) {

        if (ae.getSource() == ab) {

            try {
                int num1 = Integer.parseInt(a.getText());
                int num2 = Integer.parseInt(b.getText());

                sum = num1 + num2;

            } catch (NumberFormatException e) {
                sum = 0;
            }

            repaint();
        }
    }
}
```

```
public void paint(Graphics g) {  
  
    g.drawString("Value 1 is : " + a.getText(), 20, 100);  
    g.drawString("Value 2 is : " + b.getText(), 20, 130);  
    g.drawString("Sum of the numbers is : " + sum, 20, 160);  
}  
}
```

**OUTPUT :**

